Elliptic polylogarithms and modular forms

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ABSTRACT. This note is a preliminary draft of work describing the construction and some applications of an equivariant construction of a holomorphic realization of elliptic polylogarithms, parameterized explicitly by Steinberg symbols. The resulting families of modular forms have connections to many arithmetic and automorphic phenomena, including toric modular forms and rigid analytic cocycles.

CONTENTS

1. Introduction 2
1.1. Acknowledgements 2
2. The Kronecker section and its properties 3
2.1. The Poincaré bundle and the Kronecker section 3
2.2. Analytic trivialization 5
3. The symbol map and its specializations 8
3.1. Definition of the universal symbol maps 8
3.2. Stabilization 9
3.3. Completions and distributions 10
3.4. Relation to classes of Kings–Sprang 12
3.5. Distribution-valued specializations 13
4. Toric modular forms 15
4.1. Degeneration to $z = 0$ and toric modular forms 15
4.2. Algebraic toric modular forms 17
5. Algebraic theta lifts 18
5.1. An analytic model 18
5.2. A regularized theta series 18
5.3. Specialization and modular forms 21

5.4. Periods over totally real tori 21

References 23

1. Introduction

In this article, we construct (Theorem 3.3) canonical Steinberg-parameterized families of theta elements on powers of elliptic curves, built from the Kronecker section of the Poincaré bundle; this generalizes our previous work for $\mathbb{G}_m$ (the "$q = 0$ degeneration").

The resulting classes have specializations related to many constructions in the number theoretic literature:

- They recover (Proposition 3.5) the Eisenstein–Kronecker group cohomology classes of Kings–Sprang [KS] for powers of elliptic curves with GL_n-action, and in fact also S-arithmetic extensions thereof.
- They give an arithmetic realization of Bergeron–Charollois–Garcia's elliptic cocycle [BCG2] and Eisenstein theta lift [BCG1] for $(\mathrm{GL}_n/\mathbb{Q}, \mathrm{GL}_2/\mathbb{Q})$; cf. Section 5.
- The distribution-valued analogues (also considered in Kings–Sprang in the CM setting) of Section 3.5 lead in particular to new constructions in the framework of Darmon–Vonk's rigid analytic cocycles [DV] for general linear groups (subject of future work; presently only the bare construction is written cf. Remark 3.6).
- They give a conceptual geometric framework for the "toric modular forms" of Borisov–Gunnells, which has intriguing connections to elliptic genera of toric varieties (just as Brion-style conical generating series in the $\mathbb{G}_m$ setting are closely connected to Todd genera of toric varieties). Using this framework and inspired again by methods of Kings–Sprang, we write down in Section 4 an algebraic analogue of the Borisov–Gunnells construction over an integral base.

We also hope, in a slightly different direction, that this framework can shed light on the framework for multiple elliptic polylogarithms developed in Brown–Levin [BL] and Fonseca–Matthes [FM], though we have not yet written this down. In particular, the equivariant/cocycle organization of such multiple polylogarithms we hope may lead to a satisfactory theory of "elliptic Aomoto polylogarithms," with close ties to recent developments relating equivariant Steinberg-module constructions to motivic conjectures in the mixed Tate setting.

1.1. **Acknowledgements.** Much thanks to Marti Roset Julia for his support during this time. In accordance with the Leiden Declaration: Claude Opus and ChatGPT Astra were used, most notably to find the formulas needed in the "analytic model" section (as with the analogous analytic calculations in our previous joint work [RX]), in addition to roles in literature search and citation generation, checking conventions and calculations, being general helpful conversation partners, etc.

2. The Kronecker section and its properties

2.1. **The Poincaré bundle and the Kronecker section.** We recall here the algebraic theory of the Poincaré bundle and its canonical Kronecker section, following the treatment of Sprang [Sp, §3].

Let E be an elliptic scheme over a base S, and $E^\vee$ its dual elliptic scheme, which is the moduli space of degree-zero relative line bundles (i.e. carrying a rigidification along their zero section) on E/S. Let $\mathcal{O}$ be a commutative ring acting by endomorphisms on E over S, and let K be its fraction field; K is either $\mathbb{Q}$ or an imaginary quadratic field. Write $a \mapsto \bar{a}$ for the identity in the former case and complex conjugation in the latter.

Write $\pi : E \to S$ for the structure map and $e : S \to E$ for the zero section. There is a canonical principal polarization $\lambda : E \xrightarrow{\sim} E^\vee$ given by sending a point P to the class of the bundle $\mathcal{O}_E([-P]-[e]) \otimes \pi^* e^* \mathcal{O}_E([-P]-[e])^{\otimes -1}$ together with the canonical trivialization at zero [Sp, (2)].

The product $E \times_S E^\vee$ is equipped with the tautological Poincaré bundle whose fiber over an T-point $x \in E^\vee(T)$ is the degree-zero line bundle on $E \times_S x$ coming from the moduli problem, for any S-scheme T; it also is universal for the same moduli problem coming from the identification of E with the dual of $E^\vee$. we also have tautological rigidifications

$$r : \mathcal{P}|_{[e]\times E^\vee} \cong \mathcal{O}_{E^\vee}, \quad s : \mathcal{P}|_{E\times[e]} \cong \mathcal{O}_E$$

whose restrictions yield the rigidifications of the individual line bundle moduli, respectively come from the same rigidifications under the symmetry between E and $E^\vee$. Following loc. cit., we will write $\mathcal{P}$ for the pullback of the Poincaré bundle to $E \times E$. Sometimes, to distinguish the two factors, we will continue to refer to the second factor as $E^\vee$, which we will always be identifying with E via λ. We use the given $\mathcal{O}$-action on both copies of E; under this identification, the dual of $[a]$ is $[\bar{a}]$.

Write p_1, p_2 for the two projections of $X := E \times_S E$, and $m : E \times E^\vee \to E$ for the multiplication map; let $\omega_{E/S}$ be the Hodge bundle of E. Let $\pi_X : X \to S$ be the projection, and write $\Delta = \ker m$, $Z_1 = \ker p_1$, $Z_2 = \ker p_2$. Then as described in loc. cit., the polarization formula above leads to a description of the pulled-back Poincaré bundle as

$$\mathcal{P} \cong m^*\mathcal{O}_E([e]) \otimes p_1^*\mathcal{O}_E([e])^{\otimes -1} \otimes p_2^*\mathcal{O}_E([e])^{\otimes -1} \otimes \pi_{E\times E}^* \omega_{E/S}^{\otimes -1} \cong \mathcal{O}_{E\times E}(\Delta - Z_1 - Z_2) \otimes \pi_{E\times E}^* \omega_{E/S}^{\otimes -1}.$$

Proposition 2.1. *We have a canonical* Kronecker section

$$s_{can} \in \Gamma(E \times_S E,\ \mathcal{P} \otimes \Omega^1_{X/E}(Z_1 + Z_2))$$

characterized as the unique section whose residue along Z_1, computed via r, is

$$\mathrm{res}_{[[e]\times E]}(s_{can}) = 1 \in \Gamma(E, \mathcal{O}_E).$$

(Here, X/E should be viewed over the second, "$E^\vee$," copy of E.[1]*) Moreover, this section vanishes along the antidiagonal Δ.*

Proof. Note that the residue along Z_1 of a section of $\mathcal{P} \otimes \Omega^1_{X/E}(Z_1 + Z_2)$ takes values in

$$(\mathcal{P} \otimes \Omega^1_{X/E})|_{Z_1} \otimes N_{Z_1/X} \cong \mathcal{P}|_{Z_1} \otimes \omega_{E/S} \otimes \omega_{E/S}^{\otimes -1} \cong \mathcal{O}_E,$$

which is what makes the residue condition coherent. (Here, N refers to the normal bundle.)

Two sections of $\mathcal{P} \otimes \Omega^1_{X/E}(Z_1 + Z_2)$ with the same residue along Z_1 differ by a section of $\mathcal{P}^1_{X/E} \otimes \Omega(Z_2)$, so it suffices to prove

$$H^0(X,\ \mathcal{P} \otimes \Omega^1_{X/E}(Z_2)) = 0,$$

which is already true for the restriction fiberwise over the second factor of E: at all non-identity points, $\mathcal{P}$ restricts to a nontrivial degree-zero line bundle on the fiber copy of E, while $\Omega^1_{X/E}(Z_2)$ restricts to be trivial.

As we previously recorded, we have an identification

$$\mathcal{P} \otimes \Omega^1_{X/E}(Z_1 + Z_2) \cong \mathcal{O}_X(\Delta).$$

Writing $1_{[e]}$ for the tautological section of $\mathcal{O}_E([e])$ with a zero of order 1 along e, we define

$$s_{can} := \frac{m^* 1_{[e]}}{p_1^* 1_{[e]} \cdot p_2^* 1_{[e]}},$$

a rational section of $\mathcal{P} \otimes \Omega^1_{X/E}$. Then $\mathrm{div}(s_{can}) = \Delta - Z_1 - Z_2$; in other words, we may view s_{can} as a regular section of $\mathcal{P} \otimes \Omega^1_{X/E}(Z_1 + Z_2)$ whose vanishing locus is exactly Δ, as desired.

Since $p_1^* 1_{[e]}$ is the tautological section of $\mathcal{O}_X(Z_1)$, adjunction identifies $\mathcal{O}_X(Z_1)|_{Z_1} \cong N_{Z_1/X}$, and the residue of s_{can} along Z_1 is given by

$$\mathrm{res}_{Z_1}(s_{can}) = \frac{m^* 1_{[e]}}{p_2^* 1_{[e]}}|_{Z_1} \in \Gamma(Z_1, (m^* \mathcal{O}_E([e]) \otimes p_2^* \mathcal{O}_E(-[e]))_{Z_1}).$$

Since $m, p_2 : X \to E$ have the same restriction to Z_1 $(e, y) \mapsto y$, this is actually equal to 1. □

Observe that for any isogeny $\varphi : E \to E$, and its corresponding dual isogeny $\varphi^\vee : E \to E$ (transported via λ), we have an essentially tautological identification $(\varphi \times 1)^* \mathcal{P} \cong (1 \times \varphi^\vee)^* \mathcal{P}$, since $\varphi^\vee : E^\vee \to E^\vee$ precisely sends the moduli of a line bundle $\mathcal{L}$ to its pullback by φ. Correspondingly, for any pair of self-isogenies φ, ψ of E, we have a canonical identification

$$\gamma_{\varphi, \psi^\vee} : (\varphi \times \psi^\vee)^* \mathcal{P} \xrightarrow{\sim} (\psi \times \varphi^\vee)^* \mathcal{P}.$$

of bundles on $E \times E$. In fact, if $\varphi : E \to E'$ has different source and target, the same formula holds for the same reason on $E \times E'$.

[1] In fact, if we take it over the first copy, we get a canonically isomorphic line bundle, as both are canonically identified with the pullback by $\pi_X^* \omega_{E/S}$. The symmetry only breaks if we interpret sections as actual differential forms when writing down things in coordinates, which we will do later.

Following Sprang [Sp, §3.3], given torsion sections $s \in E[\varphi]$ and $t \in E[\psi^\vee]$, and their corresponding translation morphisms on E, we may then define a translation operator $\mathcal{U}_{s,t}^{\varphi,\psi^\vee}$ as the composite

$$(T_s \times T_t)^*(\psi \times \varphi^\vee)^*\mathcal{P} \xrightarrow{(T_s \times T_t)^*\gamma_{\psi,\varphi^\vee}} (T_s \times T_t)^*(\varphi \times \psi^\vee)^*\mathcal{P} = (\varphi \times \psi^\vee)^*\mathcal{P} \xrightarrow{\gamma_{\varphi,\psi^\vee}} (\varphi \times \psi^\vee)^\vee\mathcal{P}.$$

The switching operators $\gamma_{\varphi,\psi^\vee}$, and the derived translation operators, satisfy the natural compatibilities under composition recorded in loc. cit.

2.2. **Analytic trivialization.** The upper half-plane $\mathcal{H}$, with coordinate τ, carries a universal framed elliptic curve $E/\mathcal{H}$ which has a standard uniformization $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$. Write $\mathrm{e}(x) := e^{2\pi i x}$, and $q := \mathrm{e}(\tau)$.

In this framework, we recall the analytic trivialization of the Poincaré bundle and its relation to the Kronecker section; this point of view is written down in related ways in [BK, §1.2] and [Sp, §4.3.1]. Unlike these sources, we phrase our results in the universal setting using the classical totally odd Jacobi theta function

$$\theta(z,\tau) := 2\, q^{1/8} \sin(\pi z) \prod_{n \geqslant 1} (1-q^n)(1-q^n e^{2\pi i z})(1-q^n e^{-2\pi i z}),$$

where $q = e^{2\pi i \tau}$. This is a meromorphic function of modular weight $1/2$, quasiperiodic for the lattice $\mathbb{Z} + \tau\mathbb{Z}$ with multiplier $-\exp(-2\pi i(z + \tau/2))$ in the τ direction.

Then the étalé space of the Poincaré bundle $\mathcal{P}$ on $E \times_{\mathcal{H}} E^\vee$ may be uniformized by $\mathbb{C} \times \mathbb{C} \times \mathbb{C}$ over $\mathcal{H}$ quotiented by the relations

$$(z,w,v) \sim (z+1,w,v) \sim (z,w+1,v) \sim (z+\tau,w,\mathrm{e}(-w)v) \sim (z,w+\tau,\mathrm{e}(-z)v).$$

To see this, it suffices to use the seesaw principle to check that the restriction to each fiber over fixed w is isomorphic to the corresponding fiber of $\mathcal{P}_{\mathcal{H}}$, and that the fiber over $z = 0$ is trivial. The latter statement is clear; the former follows from the computation that $\mathcal{O}([-w] - [e])$ has multiplier system 1 and $\exp(-2\pi i w)$ along 1 and τ respectively, due to the existence of the meromorphic function $\theta(z+w)/\theta(\tau)$ with the prescribed divisor and the specified multiplier system.

Then we may define the Kronecker theta series in these coordinates

$$\Theta(z,w,\tau) := \frac{\theta'(0,\tau)\theta(z+w,\tau)}{\theta(z,\tau)\theta(w,\tau)}$$

(where the indicated derivative is in the first coordinate) is precisely a meromorphic section s_{can} of $\mathcal{P}_{\mathcal{H}}$ with divisor $\Delta - Z_1 - Z_2$, whose residue along $z = 0$ is 1.[2] This function has q-expansion

$$\Theta(z,w,\tau) = -2\pi i \left(\frac{x}{1-x} + \frac{1}{1-y} + \sum_{m,n>0} (x^m y^n - x^{-m} y^{-n}) q^{mn} \right)$$

[2]This is the same as what Brown–Levin call F in [BL, §3.4]. They use the theta function $(z,\tau) \mapsto \theta_{11}(z,\tau)/\eta(\tau)$ instead, but the eta factors cancel in the quotient. On the other hand, our Θ differs by a fiberwise-constant normalizing factor from the version in [BK, §1.2] and [Sp, §3.1 and §4.3.1]. Their definition does not vary holomorphically in τ and so cannot be used for our applications to modular forms.

where $x = \mathrm{e}(z)$, $y = \mathrm{e}(w)$.

Now, $\mathrm{SL}_2(\mathbb{Z}) \ltimes (\mathbb{Z}^2 \times \mathbb{Z}^2)$, where the semidirect product action of $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ on a vector $v \in \mathbb{Z}^2$ in either factor is by $v \mapsto v\gamma^{-1}$, acts on $\mathcal{H} \times \mathbb{C} \times \mathbb{C} \times \mathbb{C}$ extending the earlier action fiberwise, by

$$(\mu, \nu) \cdot (\tau, z, w, v) = (\tau, z + m\tau + n, w + p\tau + q, \mathrm{e}\left(-mw - pz - mp\tau\right) v),$$
$$\gamma \cdot (\tau, z, w, v) = \left(\gamma\tau, \frac{z}{c\tau + d}, \frac{w}{c\tau + d}, \mathrm{e}\left(\frac{czw}{c\tau + d}\right) v\right).$$

For Γ torsion-free, this uniformizes a line bundle on the squared universal elliptic curve over the complex modular curve $\Gamma\backslash\mathcal{H}$, and this uniformization yields the Poincaré bundle on each fiber. By the seesaw principle, it hence must be the Poincaré bundle globally tensor a line bundle pulled back from $\Gamma\backslash\mathcal{H}$; restricting to $z = w = 0$, we find that this twist is trivial.

Using the weight-$1/2$ modularity of θ, we may then compute that Θ has modular weight 1 for this action, in the sense that for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$,

$$\Theta\left(\frac{z}{c\tau + d}, \frac{w}{c\tau + d}, \gamma\tau\right) = (c\tau + d)\mathrm{e}\left(\frac{czw}{c\tau + d}\right)\Theta(z, w, \tau).$$

Hence our Θ precisely gives the section $s_{can} \in \Gamma(E \times_S E, \mathcal{P} \otimes \pi_X^* \omega_{E/S})$ for $S = \Gamma\backslash\mathcal{H}$.

Now, fix an integer $n \geqslant 1$, and let $A := E \times_S \times \ldots \times_S E$ be the n-fold self-product over S with its natural left action of $M_n(\mathcal{O})$ by endomorphisms. Set also $X_A := A \times_S A$ where we will often think of the latter as the dual abelian variety via $\lambda^{\times n}$).

We may write $\pi_i : A \to E$ for the projection to the ith elliptic coordinate. More generally, for any surjective linear forms $\ell, r : \mathcal{O}^n \to \mathcal{O}$, we have corresponding projections $\pi_\ell, \pi_r : A \to E$, and (also with the same notation) $\pi_e ll \times \pi_r : X_A \to X$. Write $\mathcal{P}_{\ell,r} := \pi_{\ell,r}^* \mathcal{P}$. Given a pair of formal symbols $\sigma := [\ell_1, \ldots, \ell_k]$, $\tau := [r_1, \ldots, r_k]$ of k projections, set

$$\mathcal{P}_{\sigma,\tau} := \mathcal{P}_{\ell_1,r_1} \otimes \ldots \otimes \mathcal{P}_{\ell_k,r_k}.$$

Note that the group law on $E^\vee$, on the level of moduli, is given by tensor product of line bundles. That is, the sum of the point on $E^\vee$ corresponding to a line bundle $\mathcal{L}_1$ on E and a point corresponding to the line bundle $\mathcal{L}_2$ is the point corresponding to the line bundle $\mathcal{L}_1 \otimes \mathcal{L}_2$. Phrased algebraically, this means that the Poincaré bundle is "bilinear" in the sense that the universal property of $\mathcal{P}$ yields a canonical isomorphism

$$m_{12}^* \mathcal{P} \cong \pi_{13}^* \mathcal{P} \otimes \pi_{23}^* \mathcal{P}$$

as bundles on $E \times_S E \times_S E$ (where we think of the third factor as the dual curve), and symmetrically

$$m_{23}^* \mathcal{P} \cong \pi_{12}^* \mathcal{P} \otimes \pi_{12}^* \mathcal{P}$$

where we think of the latter two factors as the dual curve. Here, the subscripts indicate which factors are being multiplied/projected onto. These concrete additivities are coherent under composition, and in particular may

be packaged together in more abstract language into the following "multilinearity" property of Poincaré bundles:

Proposition 2.2. *If we have a pair of symbols $\sigma = [\ell_1, \dots, \ell_k], \tau = [r_1, \dots, r_k]$ as above, and another pair $\sigma' = [\ell'_1, \dots, \ell'_k]$ and $\tau' = [r'_1, \dots, r'_k]$ with*

$$\xi := \sum_{i=1}^{k} \ell_i \otimes \bar{r}_i = \sum_{j=1}^{k'} \ell'_j \otimes \overline{r'_j} \in \mathcal{O}^n \otimes_{\mathcal{O}} \mathcal{O}^n,$$

then we have a canonical isomorphism $\mathcal{P}_{\sigma,\tau} \cong \mathcal{P}_{\sigma',\tau'}$ of bundles on X_A. We thus may justifiably label this bundle $\mathcal{P}_\xi$.

Proof. The point is that we have the Poincaré functoriality

$$\mathcal{P}_{\ell,r} \cong ((r^\vee \circ \ell) \times 1)^* \mathcal{P}_A$$

similar to what we saw before. Then there is a monoidal map from the $\mathrm{End}(A)$ to the Picard groupoid $\mathscr{P}(X)$ of line bundles on X

$$f \mapsto (f \times 1)^* \mathcal{P}_A$$

(cf. [SGA7, I, Exposé VIII, (3.2.2)]) and our claimed map is the precomposition with $\mathcal{O}^n \otimes_{\mathcal{O}} \mathcal{O}^n \to \mathrm{End}(A)$, $\ell \otimes \bar{r} \mapsto r^\vee \circ \ell = {}^t\bar{r}\,\ell$, by the preceding computation. □

Write $e_1, \dots, e_n$ for the standard basis of $\mathcal{O}^n$, and write $\xi_0^{(n)}$ for the standard orthogonal tensor

$$e_1 \otimes e_1 + \dots + e_n \otimes e_n \in \mathcal{O}^n \otimes_{\mathcal{O}} \mathcal{O}^n.$$

We will sometimes simply drop the $\xi_0^{(n)}$ from the subscript of the corresponding Poincaré bundle going forward, or replace it by the underlying abelian variety A, when clear from context.

When $n = 2$, we have the classical *Fay relation* among three of these tensor-product Poincaré bundles, which we record below in our language with the standard residue-theoretic proof.

Theorem 2.3 (Fay relation)**.** *We have the equality of meromorphic sections of $\mathcal{P}_{\xi_0^{(2)}} \otimes \pi_{X_A}^* \omega_{E/S}^{\otimes 2}$*

$$\pi_{e_1,e_1}^* s_{can} \otimes \pi_{e_2,e_2}^* s_{can} - \pi_{e_1,(1,1)}^* s_{can} \otimes \pi_{(-1,1),e_2}^* s_{can} - \pi_{e_2,(1,1)}^* s_{can} \otimes \pi_{(1,-1),e_1}^* s_{can} = 0.$$

Complex-analytically, this is the classical analytic equality (omitting τ from the notation)

$$\Theta(z_1, w_1)\Theta(z_2, w_2) - \Theta(z_1, w_1 + w_2)\Theta(z_2 - z_1, w_2) - \Theta(z_2, w_1 + w_2)\Theta(z_1 - z_2, w_1) = 0.$$

Proof. As we have already seen in a previous argument, $\mathcal{P}_{\xi_0^{(2)}} \otimes \pi_{X_A}^* \omega_{E/S}^{\otimes 2}$ has no nonvanishing global sections; thus, it suffices to show that the left-hand side is a holomorphic section, which can be quickly checked at each of the potential polar/zero divisors: for example, along $\{z_1 = 0\}$, only the first two terms contribute, and the residues are $\Theta(z_2, w_2)$, respectively its negative. All the other computations are similar. □

Trivially, we can thereby derive a relation on $\mathcal{P}_{\xi_0^{(n)}} \otimes \omega_{E/S}^{\otimes n}$ by tensoring with $(\pi_{e_3,e_3}^* \boxtimes \ldots \boxtimes \pi_{e_n,e_n}^*) s_{can}$.

3. The symbol map and its specializations

3.1. **Definition of the universal symbol maps.** We can derive a larger family of "triangle relations" by moving the Fay relation (and its tensor up to rank n) around by a certain "pushforward" action of $\mathrm{GL}_n(K) \cap M_n(\mathcal{O})$, which we now define.

Let $g \in M_n(\mathcal{O}) \cap \mathrm{GL}_n(K)$. Both g and ${}^t\bar{g}$ act on A by isogenies, and we consider

$$g \times 1 : X_A \to X_A, \quad 1 \times {}^t\bar{g} : X_A \to X_A.$$

As before, we have a canonical isomorphism

$$\gamma_g : \left(1 \times {}^t\bar{g}\right)^* \mathcal{P}_{\xi_0^{(n)}} \xrightarrow{\sim} (g \times 1)^* \mathcal{P}_{\xi_0^{(n)}}$$

from the universal property of the Poincaré bundle; here we use the fact that ${}^t\bar{g}$ is identified with $g^\vee$ under the factor-wise principal polarization of A.

Definition 3.1. *For $g \in M_n(\mathcal{O}) \cap \mathrm{GL}_n(K)$ we set*

$$\mathcal{S}_g := \mathrm{tr}_{g\times 1} \circ \ \gamma_g \circ \left(1 \times {}^t\bar{g}\right)^*,$$

where tr *denotes the trace of the finite locally free morphism $g \times 1$, combined with the projection formula. It is an endomorphism of*

$$\Gamma\left(X_A,\ \mathcal{P}_{\xi_0^{(n)}} \otimes \pi_{X_A}^* \omega_{E/S}^{\otimes n}\right)^{\text{meromorphic}}.$$

For $g \in \mathrm{GL}_n(\mathcal{O})$ the morphism $g \times 1$ is an isomorphism and $\mathcal{S}_g$ is just the pullback along $g^{-1} \times {}^t\bar{g}$.

Proposition 3.2. *The preceding definition gives a left action; i.e., $\mathcal{S}_g \circ \mathcal{S}_h = \mathcal{S}_{gh}$, and on symbols*

$$\mathcal{S}_g\left(\pi_{\ell_1,r_1}^* s_{can} \otimes \ldots \otimes \pi_{\ell_n,r_n}^* s_{can}\right) = \pi_{\ell_1 g^{-1},r_1 {}^t\bar{g}}^* s_{can} \otimes \ldots \otimes \pi_{\ell_n g^{-1},r_n {}^t\bar{g}}^* s_{can},$$

whenever the indicated vectors are all integral. In particular the scalar matrices act trivially.

Proof. The morphisms $g \times 1$ and $1 \times {}^t\bar{h}$ act on different factors of X_A and the resulting square is cartesian, so the trace along the former commutes with pullback along the latter by flat base change. Multiplicativity then follows from the coherence of the γ's together with $(gh) \times 1 = (g \times 1) \circ (h \times 1)$ and $1 \times {}^t\overline{gh} = \left(1 \times {}^t\bar{h}\right) \circ (1 \times {}^t\bar{g})$. For the effect on symbols, pullback along $1 \times {}^t\bar{g}$ replaces r_i by $r_i {}^t\bar{g}$ and the trace along $g \times 1$ divides the ℓ_i by g; since $\xi_0^{(n)}$ is preserved under $\xi \mapsto {}^t(g^{-1})\xi\, {}^t g$, the result again lies in the same space. □

Complex-analytically, the scalar distribution relation on each elliptic curve factor is the identity

$$\sum_{m,k=0}^{a-1} \mathrm{e}(kw)\,\Theta\left(\frac{z+m+k\tau}{a}, aw\right)\,d(z/a) = \Theta(z,w)\,dz,$$

where both sides are sections of $\mathcal{L}_w$ with a single simple pole at the identity with residue 1; the factor $1/a$ comes from the pullback of the invariant differential dz. This is a special case of the distribution relation treated in [Sp, Appendix A].

For a basis $[v_1, v_2, \dots, v_n]$ of a finite index subgroup of $\mathcal{O}^n$, with dual basis $[v_1^\vee, \dots, v_n^\vee]$, write

$$\Psi([v_1, v_2, \dots, v_n]) := \pi^*_{v_1^\vee, \bar{v}_1} s_{can} \otimes \pi^*_{v_2^\vee, \bar{v}_2} s_{can} \otimes \pi^*_{v_3^\vee, \bar{v}_3} s_{can} \otimes \dots \otimes \pi^*_{v_n^\vee, \bar{v}_n} s_{can},$$

interpreted as above when the dual basis is not integral; note that the tensor indexing this section is $\xi_0^{(n)}$, so it lives in the same space as the standard symbol. Applying $\mathcal{S}_g$ to the Fay relation, for

$$g = \begin{pmatrix} v_1 & \dots v_n \end{pmatrix}$$

gives then the Bykovskii relation

$$\Psi([v_1, v_2, v_3, \dots, v_n]) - \Psi([v_1, v_1+v_2, \dots, v_n]) - \Psi([v_2, v_1+v_2, \dots, v_n]) = 0$$

between meromorphic sections of $\mathcal{P}_{\xi_0^{(n)}} \otimes \pi_{X_A}^* \omega_{E/S}^{\otimes n}$. Since the scalar matrices act trivially, Ξ depends only on the lines spanned by the vectors v_i. We thus obtain a modular symbol:

Theorem 3.3. *The assignment* $[v_1, \dots, v_n] \mapsto \Psi([v_1, \dots, v_n])$ *yields a* $\mathrm{GL}_n(K)$*-equivariant modular symbol*

$$\Psi : \mathrm{St}_n(K) \to \Gamma(X_A,\ \mathcal{P}_{\xi_0^{(n)}} \otimes \Omega^{\otimes n}_{X_A/A})^{\mathrm{mero}}$$

where $g \in \mathrm{GL}_n(K)$ *acts on the target via* $\mathcal{S}_g$.

3.2. **Stabilization.** We wish to be able to specialize the symbol Ψ at torsion points of X_A. For this, we need to use the identity rigidifications of the Poincaré bundle, or rather torsion translates thereof.

From the explicit formulas defining Ψ, we observe that at a Steinberg symbol $[v_1, \dots, v_n]$, with corresponding dual basis $v_1^\vee, \dots, v_n^\vee$ its poles are along the hyperplanes

$$\ker\langle -, v_i^\vee \rangle \subset A$$

on A, and along the hyperplanes

$$\ker\langle -, \bar{v}_i \rangle \subset A^\vee$$

in the dual factor $A^\vee$.

We recall from [X2b] the *restricted* Steinberg modules: given $\mathfrak{b}, \mathfrak{c}$ proper nonzero ideals of $\mathcal{O}$ with $\mathfrak{b} + \bar{\mathfrak{c}} = \mathcal{O}$, ℓ a line of $\mathfrak{b}$-torsion in A, and λ a hyperplane modulo $\bar{\mathfrak{c}}$ (corresponding to a line of $\mathfrak{c}$-torsion in $A^\vee$ via the

conjugate pairing), then we can form

$$ {}_\lambda\mathrm{St}(n), {}^\ell\mathrm{St}(n), {}^\ell_\lambda\mathrm{St}(n) \subset \mathrm{St}(n) $$

as the submodules generated by symbols whose vectors do not span lines in the hyperplane λ mod $\bar{\mathfrak{c}}$, respectively do not span hyperplanes containing ℓ mod $\mathfrak{b}$, respectively both. The point is that if $P := (\alpha, \beta) \in A[\mathfrak{b}] \times A^\vee[\mathfrak{c}]$ is a section defined over S, with α spanning the line ℓ and β defining the hyperplane λ by $\langle \bar{v}, \beta \rangle = 0$, then $\Psi|_{{}^\ell_\lambda\mathrm{St}(n)}$ is regular at P. Note the appearance of complex conjugate in the CM case, necessary because of the use of the unitary pairing in the dual condition. (In practice, we generally choose $\mathfrak{c}$ to be an rational ideal anyway, making this moot.)

We also may set

$$ {}_{\bar{\mathfrak{c}}}\mathrm{St}(n) := \bigoplus_\lambda {}_\lambda\mathrm{St}(n), {}^{\mathfrak{b}}\mathrm{St}(n) := \bigoplus_\ell {}^\ell\mathrm{St}(n), {}^{\mathfrak{b}}_{\bar{\mathfrak{c}}}\mathrm{St}(n) := \bigoplus_{\lambda,\ell} {}^\ell_\lambda\mathrm{St}(n). $$

for $\mathrm{GL}_n(\mathcal{O})$-invariant versions.

Naively, we obtain a pullback

$$ (\alpha, \beta)^*\Psi|_{{}^\ell_\lambda\mathrm{St}(n)} : {}^\ell_\lambda\mathrm{St}(n) \to \Gamma(S, L_{\alpha,\beta} \otimes \omega_{E/S}^{\otimes n}) $$

where $L_{\alpha,\beta} := (\alpha, \beta)^*\mathcal{P}_{\xi_0^{(n)}}$.

Using the coprimality of $\mathfrak{b}$ and $\bar{\mathfrak{c}}$, we can canonically trivialize these $L_{\alpha,\beta}$. Choose $a \in \mathfrak{b}$ and $d \in \bar{\mathfrak{c}}$ with $a + d = 1$. Then $a\alpha = 0$ and $\bar{d}\beta = 0$, so $\bar{a}\beta = \beta$. Then the Poincaré rigidifications gives the sequence of identifications

$$ L_{\alpha,\beta} = L_{\alpha,\bar{a}\beta} \cong L_{a\alpha,\beta} = L_{0,\beta} \cong \mathcal{O}_S. $$

One may check using the biadditive structure of $\mathcal{P}$ that this trivialization is independent of the choice of a, d. We hence obtain a canonical specialization

$$ \Psi_{\alpha,\beta} : {}^\ell_\lambda\mathrm{St}(n) \to \Gamma(S, \omega_{E/S}^{\otimes n}). $$

functorial for base change in S.

3.3. **Completions and distributions.** Following [KS, §5], we would like to assemble our specializations into distributions. At present, we do so only in the ordinary setting for simplicity, though with a more technical treatment (as subsequent work on p-adic L-functions of Kings–Sprang has shown) we expect this assumption to be unnecessary.

Let $c > 1$ be invertible on S, and let $\beta \in A^\vee[c](S)$ have dual hyperplane λ as before. Write $\widehat{A^\vee}$ for the formal completion at the identity, and $\widehat{A^\vee}_\beta$ for the same at β. Translation yields an identification

$$ T_\beta : \widehat{A^\vee} \xrightarrow{\sim} \widehat{A^\vee}_\beta. $$

For each symbol $\sigma = [v_1, \ldots, v_n] \in {}_\lambda\mathrm{St}(n)$, observe that $(1 \times T_\beta)^*\Psi(\sigma)$ is regular in the second coordinate at zero.

If we write

$$\mathcal{L}_\beta := (1\times\beta)^*\mathcal{P}_{\xi_0^{(n)}},\ \widehat{\mathcal{P}} := \mathcal{P}_A|_{A\times\widehat{A^\vee}},\ \widehat{\mathcal{P}}_\beta := (1\times T_\beta)^*\mathcal{P}_A|_{A\times\widehat{A^\vee}},$$

these satisfy by the usual additivity $\widehat{\mathcal{P}}_\beta \cong \widehat{\mathcal{P}}\otimes \mathrm{pr}_A^*\mathcal{L}_\beta$.

We obtain by restriction of the β-translation of Ψ a formal symbol

$$\widehat{\Psi}_\beta : {}_\lambda\mathrm{St}(n) \to \Gamma\left(A\times\widehat{A^\vee}, \widehat{\mathcal{P}}_\beta\otimes\pi^*\omega_{E/S}^{\otimes n}\right)^{\text{meromorphic}},$$

where the superscript means possibly meromorphic in the first coordinate only, whose specialization to $e^\vee\in\widehat{A^\vee}$ is $(1\times\beta)^*\Psi$. If we also specialize to a torsion section α of order prime to c and the symbols to the submodule ${}^\ell_\lambda\mathrm{St}(n)$ avoiding α, then this trivializes to our previous specialization $\Psi_{\alpha,\beta}$.

We may use duality to express the values of $\widehat{\Psi}_\beta$ in terms of a character stabilization in the first factor: for this, we first pass to a cover over which the étale group scheme of c-torsion sections capitulates into being constant. The previously-defined isomorphism $\gamma_c : (1\times[c])^*\mathcal{P}_A \cong ([c]\times 1)^*\mathcal{P}_A$. For a symbol σ, put

$$\eta_c(\sigma) := \gamma_c(1\times[c]^*\Psi(\sigma).$$

For $s\in A[c](S)$, as in [Sp, §3.3], the equality $([c]\times 1)\circ(T_s\times 1) = [c]\times 1$ gives a translation operator $\mathcal{U}_s^{(c)}$ on sections of $(1\times[c])^*\mathcal{P}$. Similarly, for $\beta\in A^\vee[c](S)$, the equality $(1\times[c])\circ(1\times T_\beta) = 1\times[c]$ yields a translation operator on $(1\times[c])^*\mathcal{P}_A$, which we may conjugate by γ_c to an operator $\mathcal{V}_\beta^{(c)}$ on $(1\times[c])^*\mathcal{P}$. We also have the Weil/Oda pairing e_c by

$$\mathcal{V}_\beta^{(c)}\mathcal{U}_s^{(c)} = e_c(s,\beta)\mathcal{U}_s^{(c)}\mathcal{V}_\beta^{(c)}.$$

Proposition 3.4. *For $\beta\in A^\vee[c](S)$ and a symbol σ, we have an equality of meromorphic $(1\times[c])^*\mathcal{P}$-valued forms*

$$\mathcal{V}_\beta^{(c)}([c]\times 1)^*\Psi(\sigma) = \sum_{s\in A[c](S)} e_c(s,\beta)\mathcal{U}_s^{(c)}\eta_c(\sigma).$$

Proof. This follows from the scalar distribution relation $\mathcal{S}_{cI_n}\Psi(\sigma) = \Psi(\sigma)$, pulled back by $[c]\times 1$, then hit with $\mathcal{V}_\beta^{(c)}$. □

For $\sigma\in{}_\lambda\mathrm{St}(n)$, observe from the left-hand side that the common value is regular in the second coordinate at zero; i.e. thus the hyperplane condition is exactly the needed condition to cancel via character stabilization the second-factor polar divisors.

Noting that $[c]$ is an automorphism of $\widehat{A^\vee}$ by invertibility of c, so we may pull back the completed identity by $1\times[c]^{-1}$ and use γ_c to identify the coefficients with $\widehat{\mathcal{P}}$; we may thus think of $\mathcal{T}_s$ for as a translation operator on $\widehat{\mathcal{P}}$-valued forms, and define

$$\widehat{\Psi}_{c,\beta}(\sigma) := \sum_{s\in A[c](S)} \chi_\beta(s)\mathcal{T}_s\Psi(\sigma).$$

The resulting form is the pullback of $\widehat{\Psi}_\beta(B)$ along

$$[c] \times [c]^{-1} : A \times \widehat{A^\vee} \to A \times \widehat{A^\vee},$$

with the described coefficient identification; in coordinates, it may be written $(z, w) \mapsto ([c]z, \beta + [c]^{-1}w)$.

3.4. **Relation to classes of Kings–Sprang.** Let C_i denote the divisor on X_A coming from inserting $E \times \{0\}$ at the ith factor, and D_i the divisor coming from inserting $\{0\} \times E$ at the ith factor. The standard symbol has poles precisely along the various C_i and D_i. The preceding character stabilization removes the second-factor poles on the corresponding restricted Steinberg module. We can then parametrize the current-valued exterior derivative along the remaining first-factor divisors, and construct a symbol map valued in the Dolbeault-type complex of [KS, §3.1].

In this way, we will also see that our symbol map yields a cocycle coming from the equivariant coherent cohomology class constructed in [KS, §2], as in [X2a] in the toric setting.

We have the Orlik–Solomon/Whitney homology-style acyclic complex of $\mathrm{GL}_n(K)$-modules

$$\mathrm{OS}(n) := (\mathrm{St}_n(K) \to \mathrm{Ind}_{\mathrm{GL}_{n-1}}^{\mathrm{GL}_n} \mathrm{St}_{n-1}(K) \to \ldots \to \mathrm{Ind}_{\mathrm{GL}_1}^{\mathrm{GL}_n} \mathrm{St}_1(K) \to \mathbb{Z})$$

whose differential sends

$$[v_1, \ldots, v_k] \mapsto \sum_{i=1}^{k} (-1)^{i-1} [v_1, \ldots, \hat{v_i}, \ldots, v_k].$$

Let ${}_c\mathrm{OS}(n)$ be the c-restricted version considered in [X2a], comprised of symbols whose lines avoid a specified mod-c hyperplane, with one summand for each hyperplane. Write $\lambda^\perp$ for the line of characters corresponding to λ, so the characters in $\lambda^\perp \backslash \{1\}$ are nontrivial on each of the torsion directions allowed by that summand. In top degree we have $\bigoplus_\lambda {}_\lambda\mathrm{St}(n)$, and the corresponding Whitney homology resolution is as in [X2b].

The stabilized forms extend, by taking their residues in the first coordinate, to a morphism of complexes ${}_c\mathrm{OS}(n)_\bullet \to \mathcal{D}^{n,n-\bullet}(A(\mathbb{C}), \widehat{\mathcal{P}})$ whose top-degree map is

$$[v_1, \ldots, v_n]_\lambda \mapsto \sum_{\chi \in \lambda^\perp \backslash \{1\}} \sum_{s \in A[c]} \chi(s) \mathcal{T}_s \Psi([v_1, \ldots, v_n]).$$

The target is the Dolbeault complex of [KS, Corollary 3.6], equipped with its connection Δ''. We retain the entire completed second variable here; restriction to $w = 0$ would instead take the constant term. The construction is equivariant for automorphisms preserving the level data, and in particular for a subgroup $\Gamma \subset \mathrm{GL}_n(\mathcal{O})$ preserving the chosen decoration.

For a single nontrivial character χ, the corresponding degree-zero torsion cycle is

$$D_\chi := \sum_{s \in A[c]} \chi(s)[-s] = \sum_{s \in A[c]} \chi(s)^{-1}[s],$$

with coefficients in a ring containing the values of χ. The sign in the support comes from using pullback by T_s: a pole at zero pulls back to a pole at $-s$. Summing over $\lambda^\perp\backslash\{1\}$ gives the cycle for the hyperplane-decorated summand.

In particular, exactly as in the toric setting, we have:

Proposition 3.5. *Let Γ be a subgroup of $\mathrm{GL}_n(\mathcal{O})$ fixing D. Then the Γ-cohomology class coming from lifting a degree-zero c-torsion cycle*

$$D \in {}_c\mathrm{OS}(n)_0 \cong \mathbb{Z}\{A[c]\}^{\deg=0}$$

in the exact complex ${}_c\mathrm{OS}(n)$ is equal to the Eisenstein–Kronecker class $EK_{\Gamma,A}(D)$ of [KS, §2].

Proof. The residue of the standard Kronecker section is 1. Its tensor products and their translates therefore give the same residue maps as the differentials in the restricted complex, with the identity rigidifications of the completed coefficient along the torsion support. Thus the class obtained by lifting D has residue equal to the canonical image of D in the localization sequence. This is the characterization of $EK_{\Gamma,A}(D)$ in [KS, Theorem 2.20 and Definition 2.21]. □

Proposition 3.4 identifies the recentered symbols with these character-stabilized symbols after the indicated isogeny pullback. Thus completing at a nonzero second-factor torsion section and stabilizing in the first factor give the same classes under that identification.

3.5. **Distribution-valued specializations.** Fix a prime p; as previously described, we now work over a p-adically complete base formal scheme on which E has ordinary reduction, and such that there is a global $\hat{\mathbb{G}}_m$-trivialization of the formal group: In the CM case, we may simply complete at a split prime; in the universal one, we work over the formal Igusa tower Ig_∞ parameterizing moduli of trivialized ordinary elliptic curves, cf. [Sp, §7].

Choose torsion sections α, β with annihilating coprime ideals $\mathfrak{b}, \mathfrak{c}$ also prime to p. We will restrict to symbols whose polar divisors avoid (α, β) on the special fiber; their realizations are then regular on the formal neighborhood of (α, β).

For formal variables z, w, we have as before a canonical biextension isomorphism

$$\mathcal{P}_{\alpha+z,\beta+w} \cong L_{\alpha,\beta} \otimes \mathcal{P}_{\alpha,w} \otimes \mathcal{P}_{z,\beta} \otimes \mathcal{P}_{z,w}.$$

where we may as before trivialize the first factor canonically, as well as the middle two using multiplication by the prime-to-p orders of α and β. The last factor may be trivialized as in [KS, Proposition 5.9], which in light of the other trivializations may be transported to the left-hand side.

By our assumptions, we have a trivialization of the ordinary formal groups

$$\widehat{A} \times \widehat{A^\vee} \cong \hat{\mathbb{G}}_m^{2n}.$$

Write R for the base ring (as in both our settings, the base is pro-affine; either a completion of a ring of integers or the functions on Ig_∞). The trivialized expansion of a symbol σ is a power series

$$F_{\alpha,\beta}(\sigma) \in R[[T_1, \dots, T_{2n}]] \otimes_R \omega_{E/S}^{\otimes n}.$$

By the Mahler isomorphism (i.e. Cartier duality for the formal multiplicative group), it corresponds to a measure $\mu_{\alpha,\beta}(\sigma)$ on $\mathbb{Z}_p^{2n}$ with values in $\omega_{E/S}^{\otimes n}$, characterized by

$$F_{\alpha,\beta}(\sigma)(T_1, \dots, T_{2n}) = \int_{\mathbb{Z}_p^{2n}} \prod_{i=1}^{2n} (1+T_i)^{u_i} \, d\mu_{\alpha,\beta}(\sigma)(u).$$

A formal p-power torsion point (s,t) corresponds in these coordinates to $(\zeta_1 - 1, \dots, \zeta_{2n} - 1)$, and hence to the finite-order character $\chi_{s,t}(u) = \prod_i \zeta_i^{u_i}$. We have

$$\int_{\mathbb{Z}_p^{2n}} \chi_{s,t}(u) \, d\mu_{\alpha,\beta}(B)(u) = F_{\alpha,\beta}(\sigma)(s,t).$$

The right-hand side is the specialization of $\Psi(\sigma)$ at $(\alpha+s, \beta+t)$, using the induced formal trivialization, which agrees with the one constructed in the previous section: indeed, write $p = \mathfrak{p}\overline{\mathfrak{p}}$, with $\mathfrak{p}$ the connected direction, and suppose s, t are killed by $\mathfrak{p}^r, \mathfrak{p}^m$, respectively. If α, β are killed by $\mathfrak{b}, \mathfrak{c}$, then

$$\mathfrak{b}\mathfrak{p}^r + \overline{\mathfrak{c}}\,\overline{\mathfrak{p}}^m = \mathcal{O}.$$

Choose $a \in \mathfrak{b}\mathfrak{p}^r$ with $\bar{a} \equiv 1 \pmod{\mathfrak{c}\mathfrak{p}^m}$. Multiplication by $\bar{a}$ is invertible on the formal dual group and fixes $\beta + t$. The formal trivialization therefore specializes to

$$L_{\alpha+s,\beta+t} \cong L_{a(\alpha+s),\beta+t} \cong \mathcal{O}_S.$$

Consequently,

$$\int_{\mathbb{Z}_p^{2n}} \chi_{s,t} \, d\mu_{\alpha,\beta}(\sigma) = \Psi_{\alpha+s,\beta+t}(\sigma).$$

Applying the same completed trivialization and Mahler transform to the character-stabilized representatives of $EK_{\Gamma,A}(D)$ gives the Kings–Sprang distribution-valued classes, under the comparison given in Proposition 3.4, cf. [KS, Theorem 5.23 and Corollary 5.25].

Remark 3.6. Using the symbol complex as in the previous section, this in particular affords us distribution-valued $(n-1)$-cocycles

$$(\gamma_1 : \dots : \gamma_n) \mapsto \mu_{\alpha,\beta}([\gamma_1 e_1, \dots, \gamma_1 e_n])$$

for p-arithmetic subgroups of $\mathrm{SL}_n(\mathbb{Q})$ and $\mathrm{SL}_n(K)$ fixing (α, β), whose arithmetic significance in the rigid analytic cocycle philosophy of [DV] we will study in continuation of the present work. In the former case, for $n = 2$, we believe that spectral decomposition will relatve the resulting constructions to those in [DPV1], whereas in the latter case, we expect "Bianchi–Eisenstein cocycles" with specializations which we hope may be related to rank-n Rubin–Stark elements for totally complex fields containing K.

4. TORIC MODULAR FORMS

4.1. **Degeneration to $z = 0$ and toric modular forms.** The products of Eisenstein series in the example formulas of the previous sections are very reminiscent of Borisov–Gunnells' toric modular forms [BG], parameterized by degree functions subordinate to complete fans. We explain this relationship by summing specializations of different symbols at different points of the dual abelian variety. Agreement of these points on common faces makes the poles in the first factor cancel, after which we can specialize at the identity. Throughout this construction we take $\mathcal{O} = \mathbb{Z}$ and restrict to complete unimodular fans.

Let N be a lattice of rank n, with a fixed ordered basis, and put $M = N^\vee$. Let Σ be a complete unimodular fan in $N_\mathbb{R}$. Thus the primitive generators of every maximal cone form a basis of N; the associated toric variety X_Σ is smooth and complete. For each ray ρ, write v_ρ for its primitive generator. Choose an integer $\ell > 1$ and a degree function $d : N_\mathbb{R} \to \mathbb{R}$, linear on each cone, such that on each ray,

$$\alpha_\rho := d(v_\rho) \in \frac{1}{\ell}\mathbb{Z}\backslash\mathbb{Z}$$

For a maximal cone σ, order its generators $v_{\sigma,1}, \dots, v_{\sigma,n}$ and let $m_{\sigma,1}, \dots, m_{\sigma,n}$ be the dual basis. Write $a_{\sigma,i} = d(v_{\sigma,i})$ and write the corresponding dual vector for the linear form

$$d_\sigma = \sum_i a_{\sigma,i} m_{\sigma,i} \in \frac{1}{\ell} M$$

which agree on shared faces by assumption.

Write now $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, $A = E_\tau \otimes_\mathbb{Z} N$ and $A^\vee = E_\tau \otimes_\mathbb{Z} M$. To σ, we associate the torsion point

$$\beta_\sigma = -d_\sigma \pmod{M + \tau M} \in A^\vee[\ell].$$

Then $\langle v_{\sigma,i}, \beta_\sigma \rangle = -a_{\sigma,i}$ is nonzero in E_τ, so the specialization of the corresponding symbol in the second factor is defined. In the analytic theta trivialization, consider

$$F_{\Sigma,d}(z,\tau) = \sum_{\sigma \in \Sigma(n)} \prod_{i=1}^{n} \Theta\big(\langle m_{\sigma,i}, z\rangle, -a_{\sigma,i}, \tau\big), \qquad z \in N \otimes_\mathbb{Z} \mathbb{C}.$$

Here $\Sigma(n)$ denotes the set of maximal cones. This is, a priori, a meromorphic function near $z = 0$.

To keep track of the differential, write $\Psi\hat{}([v_1, \dots, v_n])$ for the top form obtained by wedging, in the indicated order, the differential factors in the definition of Ψ. Set

$$\epsilon_\sigma = \det(v_{\sigma,1}, \dots, v_{\sigma,n}) \in \{1, -1\},$$

where the determinant is relative to our fixed basis of N. Since

$$\bigwedge_i d\langle m_{\sigma,i}, z\rangle = \epsilon_\sigma \, dz_1 \wedge \cdots \wedge dz_n,$$

the theta expression for the signed form

$$\epsilon_\sigma \Psi^\wedge([v_{\sigma,1},\dots,v_{\sigma,n}])|_{\beta_\sigma}$$

is precisely the σ-summand of $F_{\Sigma,d}$ times $dz_1 \wedge \cdots \wedge dz_n$. Thus the cone contribution is independent of the ordering of its generators.

Proposition 4.1. *The function $F_{\Sigma,d}(z,\tau)$ is holomorphic near $z=0$.*

Proof. All possible poles near the identity are simple poles along the complexifications of the hyperplanes spanned by facets of Σ. Each facet belongs to exactly two maximal cones. Order their generators as

$$\sigma = \langle v_1,\dots,v_{n-1},v_n\rangle,\ \sigma' = \langle v_1,\dots,v_{n-1},v_n'\rangle.$$

Since both cones are unimodular and lie on opposite sides of their common facet, their dual coordinates satisfy, for $i<n$, $x_n' = -x_n, x_i'|_{x_n=0} = x_i|_{x_n=0}$ The degree values on the common generators also agree by assumption. Then the expansion shows that

$$\Theta(x,w,\tau) = \frac{1}{x} + \frac{\theta'(w,\tau)}{\theta(w,\tau)} + O(x),$$

the coefficients of $1/x_n$ in the two cone contributions are

$$\prod_{i<n}\Theta(x_i,-d(v_i),\tau),\ -\prod_{i<n}\Theta(x_i,-d(v_i),\tau)$$

respectively, which cancel; applying this to every potential hyperplane pole, we conclude holomorphicity at $z=0$. □

Proposition 4.2. *In the normalization of [BG], the toric modular form attached to (N,d) is*

$$f_{N,d}(\tau) = (2\pi i)^{-n} F_{\Sigma,d}(0,\tau).$$

It is a holomorphic modular form of weight n for $\Gamma_1(\ell)$.

Proof. Write D_ρ for the cohomology class of the invariant divisor corresponding to ρ. By [BG, Theorem 3.4],

$$f_{N,d}(\tau) = \int_{X_\Sigma} \prod_{\rho\in\Sigma(1)} \frac{D_\rho}{2\pi i}\Theta\left(\frac{D_\rho}{2\pi i}, -\alpha_\rho, \tau\right).$$

We may compute this using the localization formula in $\mathbb{G}_m$-equivariant cohomology; at the fixed point P_σ corresponding to a maximal cone σ, the contribution is the restriction of the integral to σ divided by the equivariant Euler class: on this restriction, the divisor classes for its rays restrict to $m_{\sigma,i}$ (whose product cancels with the Euler class) and all other factors restrict to 1, since $u\Theta(u,w)|_{u=0} = 1$. Then evaluation the

equivariant parameter at $2\pi i z$ gives

$$(2\pi i)^{-n} \prod_i \Theta\big(\langle m_{\sigma,i}, z\rangle, -a_{\sigma,i}, \tau\big)$$

Thus, the total evaluation is the (regular, as we have seen) value at $z = 0$. □

We remark that the independence of Borisov–Gunnells' construction to subdividing a fan is preicsely the Fay relation, the ingredient giving the Bykovskii relation needed for our modular symbol.

4.2. **Algebraic toric modular forms.** The theta trivialization used above has an algebraic replacement after completing the first factor at the identity and specializing the second factor at torsion of invertible order. Let E/S be an elliptic scheme, let ℓ be invertible on S, and choose a section $t \in E[\ell](S)$ of exact order ℓ. We retain the lattice, fan, and degree function notation from before, and write $A = E \otimes_{\mathbb{Z}} N$. Write $\widehat{A}$ for its completion along the identity, and $\mathcal{P}_A$ for the Poincaré bundle on $A \times_S A^\vee$ with its identity rigidifications.

Similar to what we did in the distributions section, we can leverage the identity rigidification into a rigidification over a formal neighborhood of a specialization:

Lemma 4.3. *For $\beta \in A^\vee[\ell](S)$, there is a canonical trivialization*

$$r_\beta : \mathcal{P}_A|_{\widehat{A}\times\{\beta\}} \xrightarrow{\sim} \mathcal{O}_{\widehat{A}}$$

extending the rigidification at the identity, compatible with base change.

Proof. Again, this is because $[\ell]$ is invertible $\widehat{A}$; at a formal coordinate z, then this is again th

$$\mathcal{P}_{A,(z,\beta)} \cong \mathcal{P}_{A,([\ell]^{-1}z,[\ell]\beta)} = \mathcal{P}_{A,([\ell]^{-1}z,0)} \cong \mathcal{O}.$$

□

Write $\alpha_\rho = k_\rho/\ell$, with $k_\rho \in \mathbb{Z}$. For every maximal cone, define $\beta_\sigma \in A^\vee[\ell](S)$ by

$$\langle v_{\sigma,i}, \beta_\sigma\rangle = -k_{\sigma,i} t.$$

The right-hand side is disjoint from the identity because $a_{\sigma,i} \notin \mathbb{Z}$. As a result, the specialization of the symbol at β_σ is regular in the second coordinate. We may then define the formal sum of meromorphic top-degree forms

$$\Omega_{\Sigma,d}(E,t) = \sum_{\sigma\in\Sigma(n)} \epsilon_\sigma\, r_{\beta_\sigma}\left(\Psi^\wedge([v_{\sigma,1},\dots,v_{\sigma,n}])|_{\widehat{A}\times\{\beta_\sigma\}}\right)$$

by using the just-established trivializations.

Proposition 4.4. *The form $\Omega_{\Sigma,d}(E,t)$ is regular on $\widehat{A}$. Its value at the identity defines, compatibly with base change, a section*

$$\mathcal{F}_{\Sigma,d}(E,t) := e^*\Omega_{\Sigma,d}(E,t) \in \Gamma(S, \omega_{E/S}^{\otimes n}),$$

where we use the fixed basis of N to identify $e^*\Omega^n_{A/S}$ with $\omega^{\otimes n}_{E/S}$. For $E = E_\tau$ and $t = 1/\ell$, we have the formula

$$\mathcal{F}_{\Sigma,d}(E_\tau, 1/\ell) = F_{\Sigma,d}(0,\tau)(dz)^{\otimes n} = (2\pi i)^n f_{N,d}\,(dz)^{\otimes n}.$$

Proof. Let F be a common facet of σ and σ', let $N_F = N \cap \mathrm{span}_{\mathbb{R}}(F)$, and write $i_F : A_F = E \otimes_{\mathbb{Z}} N_F \hookrightarrow A$. Since the degree functions agree on F, the cancellation we need amounts to

$$i_F^\vee \beta_\sigma = i_F^\vee \beta_{\sigma'}.$$

as both restricted coefficient bundles are identified with $\mathcal{P}_{A_F}|_{\widehat{A}_F \times \{i_F^\vee \beta_\sigma\}}$, compatibly trivialized by Lemma 4.3. The residue of s_{can} is 1, so the residue forms agree under this identification, and the signs ϵ_σ and $\epsilon_{\sigma'}$ cancel; regularity follows.

On E_τ, the points β_σ have representatives in $\frac{1}{\ell}M$. The theta trivialization agrees with the formal trivialization r_{β_σ}, since this is true at the identity. Thus the analytic realization of $\Omega_{\Sigma,d}$ is $F_{\Sigma,d}(z,\tau)\,dz_1 \wedge \cdots \wedge dz_n$. □

5. Algebraic theta lifts

5.1. **An analytic model.** In this section, we construct a canonical analytic representative of the degree-zero stabilized, augmented part of our equivariant coherent class when $\mathcal{O} = \mathbb{Z}$, similar to [KS, §3] (whose analytic representative already partially addresses the CM case of our construction - namely, its restriction to maximal tori). We follow the same construction as for algebraic tori in our joint work [RX], replacing the Mathai–Quillen transgression form of [BCG1] and [BCG2] by a form adapted to the holomorphic structure. The resulting representative directly computes coherent specializations, and has periods along tori associated to totally real fields which are diagonal restrictions of Hilbert–Eisenstein series (as in [BCG1]).

As before, write $A = E^n$ for the power of the universal elliptic curve over a modular curve Y with sufficiently fine level structure, and let

$$D = \sum_\xi a_\xi [\xi], \qquad \sum_\xi a_\xi = 0,$$

be a torsion cycle fixed by $\Gamma \leqslant \mathrm{SL}_n(\mathbb{Z})$. Choose $c > 1$ annihilating its support and write $A^\circ = A - A[c]$. By the preceding construction, the exterior products of our stabilized symbols represent $EK_{\Gamma,A}(D)$. Denote its image under the augmentation $\widehat{\mathcal{P}} \to \mathcal{O}_A$ by

$$\kappa_D \in H^{n-1}_\Gamma(A^\circ, \Omega^n_{A/Y}).$$

This is the weight-zero quotient of the completed coefficient. All the constructions below are linear in D, so we may also use cycles with character values as coefficients.

5.2. **A regularized theta series.** Let $V = \mathbb{C}^n$ and $X = \mathrm{GL}_n(\mathbb{R})/\mathrm{O}(n)$. Identify X with the positive-definite real symmetric matrices, with action

$$g \cdot (H, z) = (g^{-t} H g^{-1}, gz).$$

The elliptic family over $\mathcal{H}$ is

$$A_{\mathcal{H}} = (\mathcal{H} \times V)/\mathbb{Z}^{2n}, \qquad (m,k)\cdot(\tau,z) = (\tau, z+m\tau+k).$$

For torsion-free Γ, the contractible space X is a model for $E\Gamma$, and the quotient $\Gamma\backslash(X \times A_{\mathcal{H}})$ computes the corresponding equivariant cohomology. (For a general arithmetic Γ, since everything is with characteristic-zero field coefficients, we may extend all results by restricting and corestricting from a torsion-free finite-index subgroup.)

As in [KS, §3.2], we use a total complex combining a distributional Dolbeault complex with coefficients in $\Omega^n_{A_{\mathcal{H}}/\mathcal{H}}$, together with a de Rham differential along X:

$$\mathcal{D}^r = \bigoplus_{a+q=r} \mathcal{A}^a_X\big(\mathcal{D}^{0,q}_{A_{\mathcal{H}}}(\Omega^n_{A_{\mathcal{H}}/\mathcal{H}})\big), \qquad d_{tot} = d_X + \bar\partial_{\mathcal{H}\times V},$$

with the usual sign conventions. Here the notation on the right means smooth X-families of currents; locally such currents are paired with compactly supported smooth test forms in the complex variables; the compactly supported test-form spaces are given the standard ind-topology. As in loc. cit., after taking Γ-invariants and descent in τ, our complex computes equivariant coherent cohomology with coefficients in $\Omega^n_{A/Y}$.

In expressions on the covering space we write the holomorphic coefficient last, and write $\omega = dz_1 \wedge \cdots \wedge dz_n$. We include the τ-direction in the antiholomorphic differential in d_{tot}.

Set $v = \operatorname{Im}\tau$ and set

$$u = \frac{H\bar z}{v}, \qquad Q = Q_{H,\tau}(z) = \frac{\bar z^t H z}{v} = z^t u, \qquad \varepsilon_n = (-1)^{n(n-1)/2}.$$

Define

$$\Omega(H,z,\tau) = \sum_{j=1}^{n} (-1)^{j-1} u_j\, d_{tot}u_1 \wedge \cdots \wedge \widehat{d_{tot}u_j} \wedge \cdots \wedge d_{tot}u_n$$

and

$$\varphi = \frac{\varepsilon_n}{(2i)^n} e^{-\pi Q} d_{tot}u_1 \wedge \cdots \wedge d_{tot}u_n \wedge \omega,$$
$$\psi = \frac{\varepsilon_n}{(2i)^n} e^{-\pi Q} \Omega(H,z,\tau) \wedge \omega.$$

One may compute that φ has integral 1 on each fiber (given the usual complex orientation), and that these are rapidly decreasing forms.

Since $d_{tot}z = 0$, we have

$$d_{tot}Q = \sum_i z_i d_{tot}u_i$$

from which we deduce $d_{tot}\varphi = 0$. For $t > 0$ let $\langle t\rangle$ denote $z \mapsto \sqrt{t}\, z$, and write $\varphi_t = \langle t\rangle^*\varphi$, $\psi_t = \langle t\rangle^*\psi$; we then compute

$$\varphi_t = \frac{\varepsilon_n}{(2i)^n} t^n e^{-\pi t Q} \bigwedge_i d_{tot}u_i \wedge \omega, \qquad d_{tot}\psi_t = t\frac{\partial}{\partial t}\varphi_t.$$

Equivalently, we have the antiholomorphic contraction formula $\psi = \iota_{\sum_i \bar{z}_i \partial/\partial \bar{z}_i} \varphi$. Then the Mellin transform at zero is

$$\eta(H, z, \tau) = \int_0^\infty \psi_t \frac{dt}{t} = \frac{\varepsilon_n (n-1)!}{(2\pi i)^n} \frac{\Omega(H, z, \tau)}{Q_{H,\tau}(z)^n} \wedge \omega,$$

smooth away from the zero section.

Lemma 5.1. *The forms φ and ψ are $\mathrm{GL}_n(\mathbb{R})$-invariant, as well as satisfying the modular invariance property*

$$(H, z, \tau) \mapsto (H, z/(c\tau + d), \gamma\tau), \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R}).$$

Further, $d_{tot}\eta = \delta_0$, the current along the zero section.

Proof. Under $g \in \mathrm{GL}_n(\mathbb{R})$, Q is unchanged and u transforms by ${}^t g^{-1}$, cancelling the action ω. For the modularity, we have

$$v' = v/|c\tau + d|^2, \qquad Q' = Q, \qquad u' = (c\tau + d)u, \qquad \omega' = (c\tau + d)^{-n}\omega.$$

Since $c\tau + d$ is holomorphic, $d_{tot}(c\tau + d) = 0$: hence both $\bigwedge_i d_{tot} u_i$ and Ω pick up the weight-n automorphy factor $(c\tau + d)^n$, which cancels the factor on ω. Finally, φ_t tends to zero locally as a current as $t \to 0$, and tends to δ_0 as $t \to \infty$, so integrating $d_{tot}\psi_t = t\partial_t \varphi_t$ gives the assertion. □

Choose lifts $\widetilde{\xi}(\tau) = a'_\xi \tau + b'_\xi$ of the torsion sections, with $a'_\xi, b'_\xi \in \frac{1}{c}\mathbb{Z}^n$. For $m, k \in \mathbb{Z}^n$, let

$$T_{\xi,m,k}(H, z, \tau) = (H, z - \widetilde{\xi}(\tau) - m\tau - k, \tau).$$

Define the theta sums

$$\theta_{\psi,t}(D) = \sum_\xi a_\xi \sum_{m,k \in \mathbb{Z}^n} T^*_{\xi,m,k} \psi_t, \qquad \theta_{\varphi,t}(D) = \sum_\xi a_\xi \sum_{m,k \in \mathbb{Z}^n} T^*_{\xi,m,k} \varphi_t.$$

All translations are holomorphic in (τ, z), so their pullbacks commute with d_{tot}; these sums are independent of the lifts, descend to $A_{\mathcal{H}}$, and converge to smooth forms for $t > 0$. We define the global Mellin trasnsform

$$E_D(s) = \int_0^\infty \theta_{\psi,t}(D) t^s \frac{dt}{t}.$$

Proposition 5.2. *Away from the support of D, $E_D(s)$ is an entire function of s with values in smooth forms. Its value at zero satisfies*

$$d_{tot} E_D(0) = \delta_D$$

For every integer $a > 0$, we have

$$[a]_* E_D(s) = a^{2s} E_{[a]_* D}(s).$$

In particular $E_D(0)$ is trace-fixed whenever $a \equiv 1 \pmod{c}$.

We conclude that $[(2\pi i)^n E_D(0)]$ represents the equivariant local cohomology class:

Corollary 5.3. *The current $\mathcal{E}_D$ represents κ_D in equivariant coherent cohomology. It is smooth on $A - \operatorname{supp} D$, so the representative extends to this larger open set.*

Fiberwise, the image of our class under the Hodge–de Rham edge map is the equivariant Eisenstein class of [BCG1], as they both are characterized by compatible residue and trace-fixedness conditions.

5.3. **Specialization and modular forms.** Let $x : Y \to A - \operatorname{supp} D$ be a torsion section fixed by Γ, passing to higher level if necessary. Restriction of the smooth Dolbeault representative along x gives a map of total complexes

$$x^* : \mathcal{A}_X^\bullet\big(\mathcal{A}^{0,\bullet}_{A-\operatorname{supp} D}(\Omega^n_{A/Y})\big) \longrightarrow \mathcal{A}_X^\bullet\big(\mathcal{A}_Y^{0,\bullet}(x^*\Omega^n_{A/Y})\big).$$

Here the antiholomorphic forms are pulled back along x, and the relative holomorphic form is pulled back as a coefficient. In particular, this map commutes with $d_X + \bar\partial_Y$. We may identify $x^*\Omega^n_{A/Y}$ canonically with $\omega_{E/Y}^{\otimes n}$.

Since Y is affine, the resulting equivariant coherent class gives

$$x^*\kappa_D \in H^{n-1}\big(\Gamma, \Gamma(Y, \omega_{E/Y}^{\otimes n})\big).$$

Integrating the components of $x^*\mathcal{E}_D$ over simplices in X gives a representative in the total complex.

Corollary 5.4. *Let C be a compact oriented $(n-1)$-cycle in $\Gamma\backslash X$. Then*

$$\Theta_{D,x,C} := \int_C (x^*\mathcal{E}_D)_{n-1,0}$$

is the cap product of $x^*\kappa_D$ with $[C]$, giving a holomorphic section of $\omega_{E/Y}^{\otimes n}$.

This yields the modular-form-valued cohomology class we seek, which is an algebraic, integral version of the theta lift of [BCG1, §13]. The lifts of maximal anisotropic tori are diagonal restrictions of Hilbert–Eisenstein series for the corresponding fields, as in loc. cit.

5.4. **Periods over totally real tori.** Let F be a totally real field of degree n, with ordered embeddings $\sigma_1, \dots, \sigma_n$, and let L be a fractional ideal of F. Choose an ordered $\mathbb{Z}$-basis $\ell_1, \dots, \ell_n$ of L such that

$$B = (\sigma_i(\ell_j))_{i,j}, \qquad \Delta_L = \det B > 0.$$

This identifies $E_\tau \otimes L$ with E_τ^n. In embedding coordinates $w = Bz$, its period lattice is

$$\{(\sigma_i(a)\tau + \sigma_i(b))_i : a, b \in L\}, \qquad \omega_F = dw_1 \wedge \cdots \wedge dw_n = \Delta_L \omega.$$

Let $U \subset \mathcal{O}_F^{\times,+}$ be a subgroup of finite index contained in Γ and fixing x and every section in the support of D. Such a subgroup exists after imposing congruences on the units. We use the embedding $U \to \mathrm{SL}_n(\mathbb{Z})$ given by multiplication on L.

Consider

$$T = \{(r_1, \dots, r_n) \in \mathbb{R}^n_{>0} : \prod_i r_i = 1\}, \qquad H(r) = B^t \operatorname{diag}(r_1, \dots, r_n) B.$$

The action induced by a unit u is $r_i \mapsto \sigma_i(u)^{-2} r_i$, so the map $T \to X$ is U-equivariant. The compact torus $U\backslash T$ maps to $\Gamma\backslash X$ and defines our cycle C_U.

To specify the orientation and the constant in the integral, set

$$R = \sum_i r_i \frac{\partial}{\partial r_i}, \qquad \nu_T = \iota_R \left(\frac{dr_1}{r_1} \wedge \cdots \wedge \frac{dr_n}{r_n} \right) \Big|_T .$$

We orient T by $\varepsilon_n \nu_T$, and C_U by the resulting quotient orientation. Equivalently,

$$\nu_T = n(-1)^{n-1} \bigwedge_{i<n} \frac{dr_i}{r_i}.$$

Write

$$x(\tau) = a_0 \tau + b_0, \qquad \xi(\tau) = a_\xi \tau + b_\xi \pmod{L\tau + L}, \qquad a_0, b_0, a_\xi, b_\xi \in F,$$

now using d_ξ for the coefficient of $[\xi]$ in D to distinguish it from the coordinate a_ξ. Set

$$\mathcal{L}_\xi = L^2 + (a_0 - a_\xi, b_0 - b_\xi) \subset F^2.$$

Each of these cosets is U-stable, and none contains $(0,0)$ because x avoids the support of D.

For $\boldsymbol{\tau} = (\tau_1, \dots, \tau_n) \in \mathcal{H}^n$ and $\operatorname{Re} s$ sufficiently large, define

$$\mathscr{E}_{L,D,x}(\boldsymbol{\tau}, s) = \sum_\xi d_\xi \sum_{(a,b) \in U\backslash \mathcal{L}_\xi} \prod_{i=1}^n \frac{(\operatorname{Im} \tau_i)^s}{(\sigma_i(a)\tau_i + \sigma_i(b)) |\sigma_i(a)\tau_i + \sigma_i(b)|^{2s}}.$$

We use the Hecke continuation in s at zero and write $\mathscr{E}_{L,D,x}(\boldsymbol{\tau}) = \mathscr{E}_{L,D,x}(\boldsymbol{\tau}, 0)$.

Theorem 5.5. *The series $\mathscr{E}_{L,D,x}(\boldsymbol{\tau}, s)$ is regular at $s = 0$, and its value there is a holomorphic Hilbert–Eisenstein series of parallel weight $(1, \dots, 1)$ for a congruence subgroup preserving the indicated data. The period of our coherent class is*

$$\big\langle x^* \kappa_D, [C_U] \big\rangle = \mathscr{E}_{L,D,x}(\tau, \dots, \tau)\, \omega_F.$$

Thus its scalar coefficient with respect to ω is $\Delta_L \mathscr{E}_{L,D,x}(\tau, \dots, \tau)$.

Proof. We compute the integral for s in the range of absolute convergence before analytically continuing to $s = 0$. Fix τ and put $v = \operatorname{Im} \tau$. For $(a, b) \in \mathcal{L}_\xi$, write

$$w_i = \sigma_i(a)\tau + \sigma_i(b), \qquad Q_r(w) = \sum_i \frac{r_i |w_i|^2}{v}.$$

Changing from the lattice basis to embedding coordinates in our kernel gives $u_i = r_i\bar{w}_i/v$ and holomorphic coefficient ω_F. The component of Ω of degree $n-1$ along T is

$$\left(\prod_i \frac{\bar{w}_i}{v}\right)\nu_T.$$

Indeed, before restriction to T it is $\prod_i(\bar{w}_i/v)\,\iota_R(dr_1\wedge\cdots\wedge dr_n)$, and $\prod_i r_i = 1$ on T.

The substitution

$$q_i = tr_i, \qquad T\times\mathbb{R}_{>0}\xrightarrow{\sim}\mathbb{R}^n_{>0}$$

satisfies

$$\bigwedge_i \frac{dq_i}{q_i} = \frac{dt}{t}\wedge\nu_T, \qquad t^{ns} = \prod_i q_i^s.$$

The sign ε_n in the kernel cancels the sign in our chosen orientation of T. Each U-orbit in the lattice sum therefore contributes

$$\frac{1}{(2i)^n}\prod_{i=1}^n\left(\int_0^\infty \frac{q_i\bar{w}_i}{v}e^{-\pi q_i|w_i|^2/v}q_i^s\,\frac{dq_i}{q_i}\right)\omega_F.$$

The one-variable integral is thus

$$\int_0^\infty \frac{q\bar{w}}{v}e^{-\pi q|w|^2/v}q^s\,\frac{dq}{q} = \frac{\Gamma(1+s)}{\pi^{1+s}}\frac{v^s}{w|w|^{2s}}.$$

Consequently,

$$\int_{C_U}(x^*E_D(ns))_{n-1,0} = \frac{\Gamma(1+s)^n}{(2\pi i)^n\pi^{ns}}\mathscr{E}_{L,D,x}(\tau,\dots,\tau,s)\,\omega_F.$$

for $\operatorname{Re} s$ large enough.

The left side is entire in s, by Proposition 5.2 and compactness of C_U. It follows that the right side is regular at zero. Multiplying by $(2\pi i)^n$ and setting $s=0$ proves the period formula by Corollary 5.4. □

A single ideal L gives the corresponding partial Hilbert–Eisenstein series. Linear combinations over ideal classes, with the same weights on the torus cycles, give the associated series associated to ideal class characters.

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